Save New Examples for Testing

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Helpful Classroom Sorter

WORKBOOK · 1 OF 21

Recall and choose your starting point

> Record your starting explanation for Save New Examples for Testing before opening notes. Show any later correction separately.

Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

Build a Human Classifier

Without opening the old lesson, explain one key idea from Build a Human Classifier. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

Debug the Cause

Without opening the old lesson, explain one key idea from Debug the Cause. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

Algorithms as Recipes

Without opening the old lesson, explain one key idea from Algorithms as Recipes. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

WORKBOOK · 2 OF 21

Practise with fading help

> Complete the supported Save New Examples for Testing trace and record the first step you can explain without help.

Use W2: Preserve the original result for Keeping final tests unseen. Record the answer once; the lesson and workbook refer to the same attempt.

Given inputs	Procedure I chose	Middle steps	Result and check

The step I completed without help, and the hint I needed

WORKBOOK · 3 OF 21

Individual reasoning check

> Check the proposed Save New Examples for Testing explanation against the supplied facts before discussing it.

Use the worked case for Keeping final tests unseen. Proposed explanation: “A new file makes old records unseen”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

My explanation before partner discussion

WORKBOOK · 4 OF 21

Independent transfer case

> Write one first attempt at the reserved independent case before feedback. Repeated copies of that case refer to the same recorded attempt.

Independent case: Run a separate ribbon experiment

Before receiving any test labels, **freeze** RibbonA: whole cm lengths 1 through 12, long at least 6 cm and short otherwise. Save **prediction**s for 5, 7, 9 and 11 cm.

Ask your teacher to reveal the reserved labels, then record agreements, disagreements and a limited result statement.

If you later choose threshold 8 after seeing these labels, explain how you must describe its result on the same records.

> **My independent explanation, trace and limits**

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

WORKBOOK · 5 OF 21

Your learning target

> Connect the Save New Examples for Testing target to a trace and a justified conclusion.

Eight cards are used to invent a rule. Four sealed cards are then opened; the team changes its rule after seeing errors. Can those four cards still be called an untouched final test? Plan what to do next.

WORKBOOK · 6 OF 21

Solve the given case

> Solve the supplied Save New Examples for Testing case and show the deciding evidence.

Cards A–F are used for practice. G and H remain sealed.

> Which can provide a fresh test, and under what condition?

Show your trace, calculation or **evidence**

> **Write or draw your reasoning and evidence.**

Check a changed case

> Apply the procedure to the changed case and record why the conclusion follows.

> You teach a sorter using cards A, B and C. You keep D and E unseen. Which cards provide the fairer first test? What changes if you show D during teaching?

My independent answer and reason

Write or draw your reasoning and evidence.

Week 15 Test and reflect

WORKBOOK · 8 OF 21

Design a revealing test

> Specify a changed input, expected result and observation separately.

Create a changed **input**, boundary case or missing-information case. Write the expected answer or safe response before testing.

Test record	My evidence
Changed input or condition	
Expected answer or fallback	
Observed result	
Why the result occurred	

WORKBOOK · 9 OF 21

Project checkpoint

> Justify what Save New Examples for Testing adds to Helpful Classroom Sorter and keep the

> evidence.

Create separate learn and test envelopes for the Classroom Sorter and label who may open each.

The project change and **evidence** I will keep

> **Write or draw your reasoning and evidence.**

Self check

> Judge explanation, evidence, transfer and limits separately; attach the work behind your judgement.

- I can explain the mechanism.
- I can show evidence.
- I can handle a changed case.
- I can name a limit.

Mark each as not yet, with help or independently, and attach supporting work.

Teacher review _____ Next step _____

Transfer practice from the original programme

> Use this earlier case as transfer practice. A case already discussed with feedback is not fresh assessment evidence.

Eight cards are used to invent a rule. Four sealed cards are then opened; the team changes its rule after seeing errors. Can those four cards still be called an untouched final test? Plan what to do next.

> **My independently reasoned answer**

Record support used: none, hint or teacher explanation.

W1 Assign the evidence its role

> Assign the evidence its role. Apply the supplied Save New Examples for Testing facts and keep your reasoning.

For each event, state whether examples are being used for development or independent evaluation of a fixed model: comparing candidates on **fitting** labels; scoring a frozen model on previously reserved examples once; changing a threshold after reading a test score.

Explain your last choice.

My reasoning and evidence

WORKBOOK · 13 OF 21

W2 Preserve the original result

> Preserve the original result. Apply the supplied Save New Examples for Testing facts and keep your reasoning.

TowerA uses threshold 5 on reserved heights 3, 5, 7 and 9 cm. The labels are short, short, tall and tall. Write its **prediction**s, agreements and disagreement.

Then state why TowerB with threshold 6, selected after seeing these labels, cannot claim four independent test successes on these same records.

My reasoning and evidence

WORKBOOK · 14 OF 21

W3 Detect a less visible leak

> Detect a less visible leak. Apply the supplied Save New Examples for Testing facts and keep your reasoning.

A team leaves reserved labels out of its **fitting** table but tries six rules and reads a test score for each. It chooses the highest-scoring rule. Explain how information still reached model selection.

Would renaming or reordering the same test records repair the problem?

My reasoning and evidence

WORKBOOK · 15 OF 21

W4 Run a separate ribbon experiment

> Attempt before feedback. If this repeats the selected independent case, reuse that first attempt; otherwise keep a separate record. Run a separate ribbon experiment. Apply the supplied Save New Examples for Testing facts and keep your reasoning.

Before receiving any test labels, **freeze** RibbonA: whole cm lengths 1 through 12, long at least 6 cm and short otherwise. Save **prediction**s for 5, 7, 9 and 11 cm.

Ask your teacher to reveal the reserved labels, then record agreements, disagreements and a limited result statement.

If you later choose threshold 8 after seeing these labels, explain how you must describe its result on the same records.

My reasoning and evidence

WORKBOOK · 16 OF 21

W5 Audit a proposed final set

> Audit a proposed final set. Apply the supplied Save New Examples for Testing facts and keep your reasoning.

A class reserves ten copies of one easy development example in a new file. The model agrees with all ten labels.

Explain two reasons this gives weak **evidence** about new cases, and propose a better final evaluation plan without promising perfect future performance.

My reasoning and evidence

WORKBOOK · 17 OF 21

Feedback, return check and learning record

> Record the first repaired step and a different case to revisit after a delay.

Evidence	My record
Worked alone / hint / model	
First step I repaired and why	
New case checked later; date and result	
What I can now explain without notes	

Evidence	My record
One remaining question	

Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

WORKBOOK · 18 OF 21

Pictorial case practice: Save New Examples for Testing

> A fictional envelope exercise evaluates a fixed pencil-length sorter. Explain the labelled step before checking the changed condition.

A fictional envelope exercise evaluates a fixed pencil-length sorter.

Freeze PencilP: long at least 6 cm; short below 6.

Reserved valid lengths are 3,5,6,9 cm. Teacher envelope labels, revealed only after **prediction**s, are short,long,long,long.

AI ACADEMY / AI-1/15

Frozen predictions versus reserved labels

	Length cm	PencilP	Reference	Result
Test 1	3	short	short	agree
Test 2	5	short	long	disagree
Test 3	6	long	long	agree
Test 4	9	long	long	agree

Read a cell using both its row and column labels.

Follow the labelled picture with the matching explanation. Open the vector diagram

AI ACADEMY / AI-1/15

Save New Examples for Testing

Start with the supplied facts
Follow each arrow. Ask which fact allows the next step.



1. Freeze
Write PencilP and its boundary before opening the label envelope.



2. Predict
For 3,5,6,9 cm predict short,short,long,long.



3. Reveal references
Envelope labels are short,long,long,long.



4. Compare
Three of four agree; the disagreement is at 5 cm.

CHANGE ONE THING
After feedback, change the boundary to 5.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

Does 4/4 on the replay replace the original 3/4 result? Explain.

My picture-based explanation and changed-case prediction

Record the help used

Support used: alone / hint / worked example. This record is practice **evidence**.

Project connection: Maintain dated sorter versions and separate **development examples** from reserved project-evaluation cases.

WORKBOOK · 19 OF 21

Project portfolio milestone

> Keep this week's input, deciding step, result and limit together in the Helpful Classroom Sorter record.

Project: Helpful Classroom Sorter

Current milestone: Test changed cases. Keep normal, boundary and failure cases; compare expected and observed results.

Maintain dated sorter versions and separate **development examples** from reserved project-evaluation cases.

Evidence to keep	My record
This week's input and deciding step	
Prediction and actual result (or labelled paper trace)	
One change since the previous checkpoint and why	
One error or limit and the next test	

Explain what this week adds to your project. Reuse last week's record rather than starting again.

WORKBOOK · 20 OF 21

Supported practice and learning evidence

> Try the supported questions before comparing with hints and feedback. Record what you repaired and the evidence you retained.

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

A fictional envelope exercise evaluates a fixed pencil-length sorter.

Practice 1: explain the deciding evidence

Which labelled step matches this detail? Three of four agree; the disagreement is at 5 cm.

1. Reveal references

2. Compare

3. **Freeze**

4. Predict

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Feedback · compare after trying

After your attempt, compare with the supported explanation: Compare Three of four agree; the disagreement is at 5 cm. The **prediction** record must exist before the labels are seen.

PencilP scores 3/4=75% agreement on these four selected cases. Moving the boundary after inspecting the envelope makes this a development exercise; the same four cases can no longer serve as untouched final evaluation.

Practice 2: explain the deciding evidence

Changed case · predict first

Changed-case practice: After feedback, change the boundary to 5. Which conclusion is supported by the supplied facts?

1. The revised rule matches all four known labels but needs a different final test.

2. We can decide the result without examining the changed **input** or the procedure.

3. This one example guarantees correct results for every future **input**.

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Feedback · compare after trying

After your attempt, compare with the supported explanation: The revised rule matches all four known labels but needs a different final test.

These labels influenced the revision, so replaying them is **evidence** of repair rather than unseen performance.

Repair a missing building block

Review Build a Human Classifier (ai-1/14). Goal: Construct a classifier with a stated **input** format, selected rule and recorded **output**.

Short practice: Trace 0 cm and explain why clarification is different from predicting short.. Return to this week and explain what you changed.

Review Debug the Cause (ai-1/12). Goal: Compare expected and observed traces to locate the first difference. Short practice: Propose **input** 0 and give both predicted **output**s without inventing an observed result..

Return to this week and explain what you changed.

Review Algorithms as Recipes (ai-1/9). Goal: Trace an ordered algorithm using the current value after every instruction. Short practice: Use the arrows to explain where the two zero-start runs first differ..

Return to this week and explain what you changed.

Keep your completed work

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

Drafts stay in this browser when storage is available; clear drafts on a shared device.

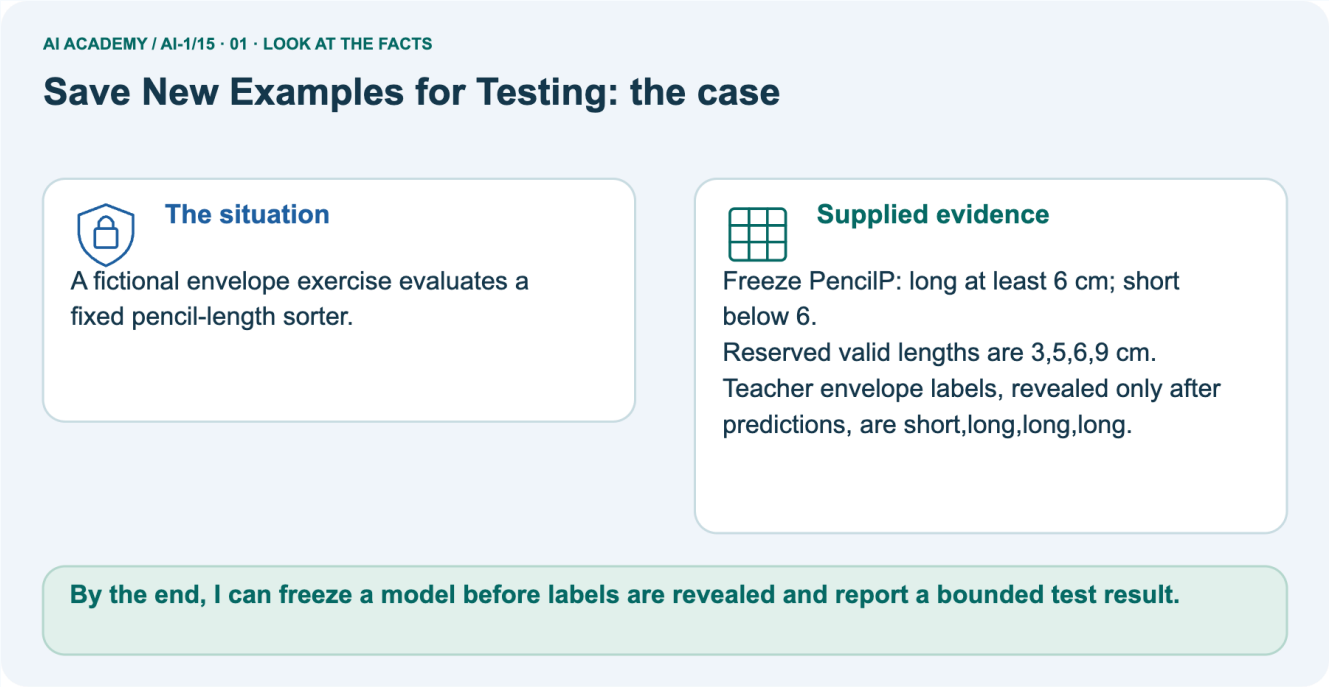
> **What I repaired, the evidence I kept and the next test**
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> | |
> | |
> | |

WORKBOOK · 21 OF 21

Visual concept journey

> Explain each Save New Examples for Testing picture from its labels, then record a relevant limit.

01. Save New Examples for Testing: the case



Follow the labelled picture with the matching explanation. Open the vector diagram

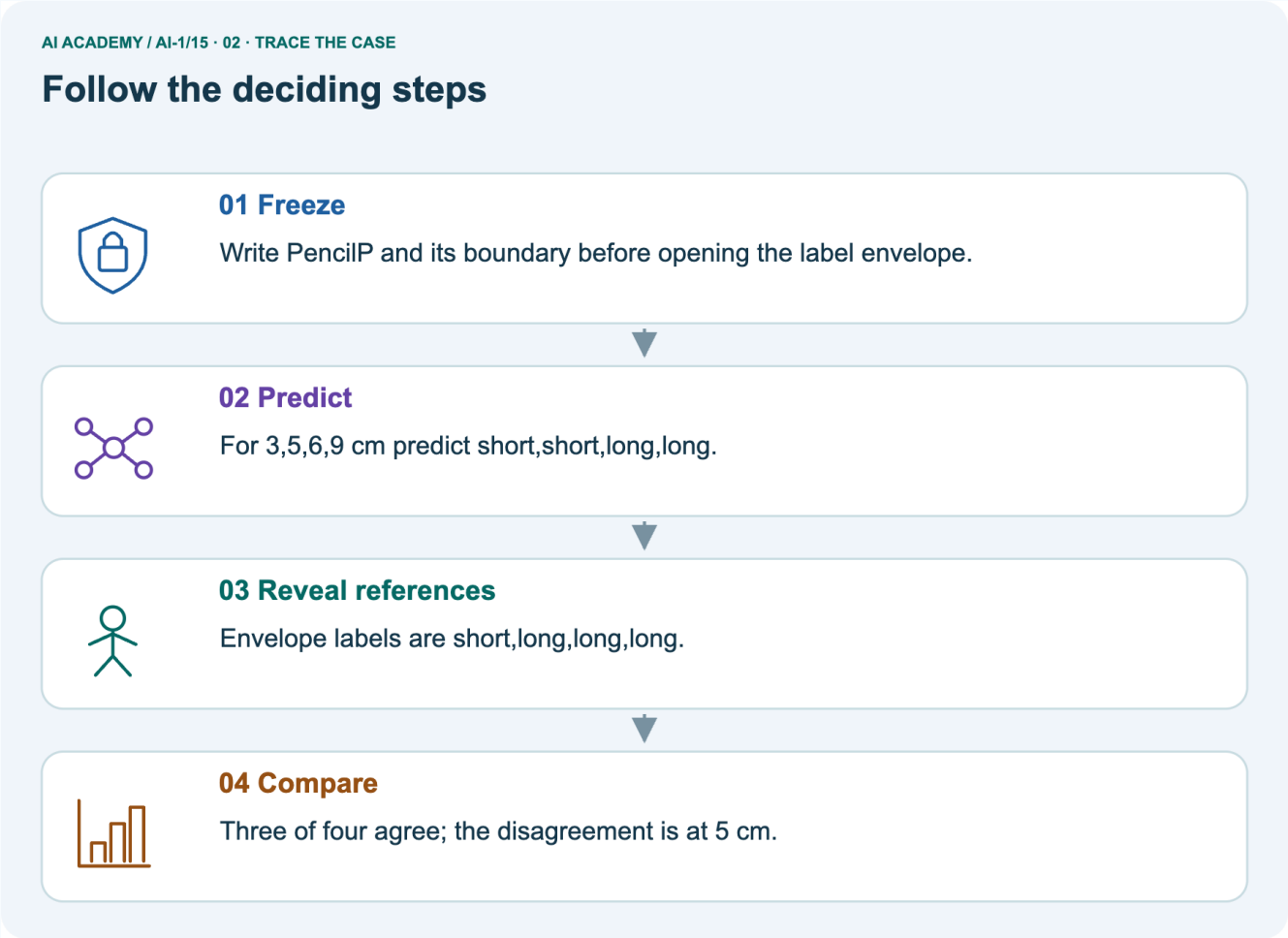
A fictional envelope exercise evaluates a fixed pencil-length sorter. Supplied facts: **Freeze** PencilP: long at least 6 cm; short below 6. Reserved valid lengths are 3,5,6,9 cm.

Teacher envelope labels, revealed only after **prediction**s, are short,long,long,long.

> Point to the given fact that matters most. Explain why it matters.

My picture explanation, deciding evidence and one limit

02. Follow the deciding steps



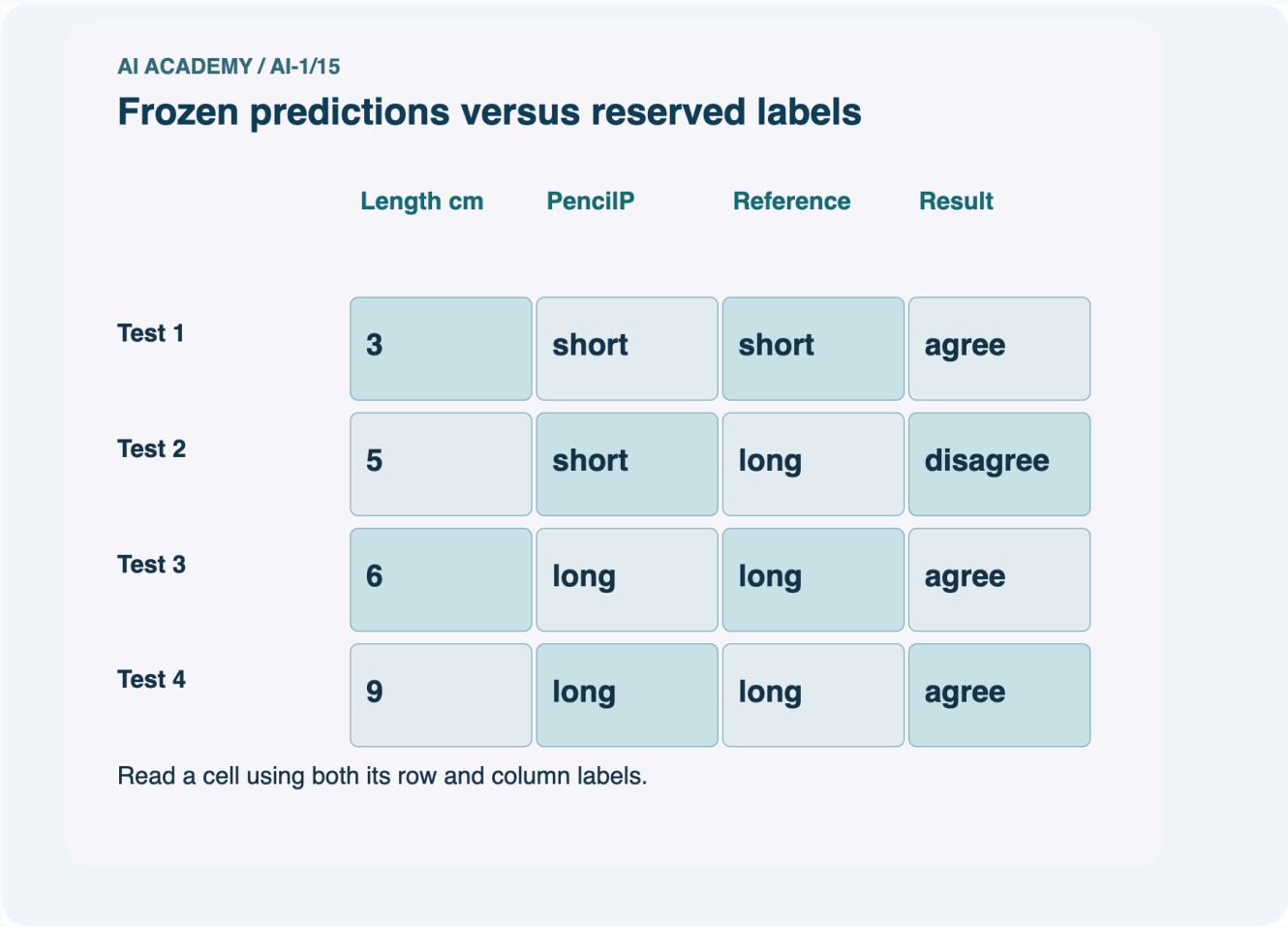
Follow the labelled picture with the matching explanation. Open the vector diagram

Freeze: Write PencilP and its boundary before opening the label envelope. → Predict: For 3,5,6,9 cm predict short,short,long,long. → Reveal references: Envelope labels are short,long,long,long. → Compare: Three of four agree; the disagreement is at 5 cm.

> Point to each numbered stage and explain which recorded **input** it uses.

My picture explanation, deciding evidence and one limit

03. Frozen predictions versus reserved labels



Follow the labelled picture with the matching explanation. Open the vector diagram

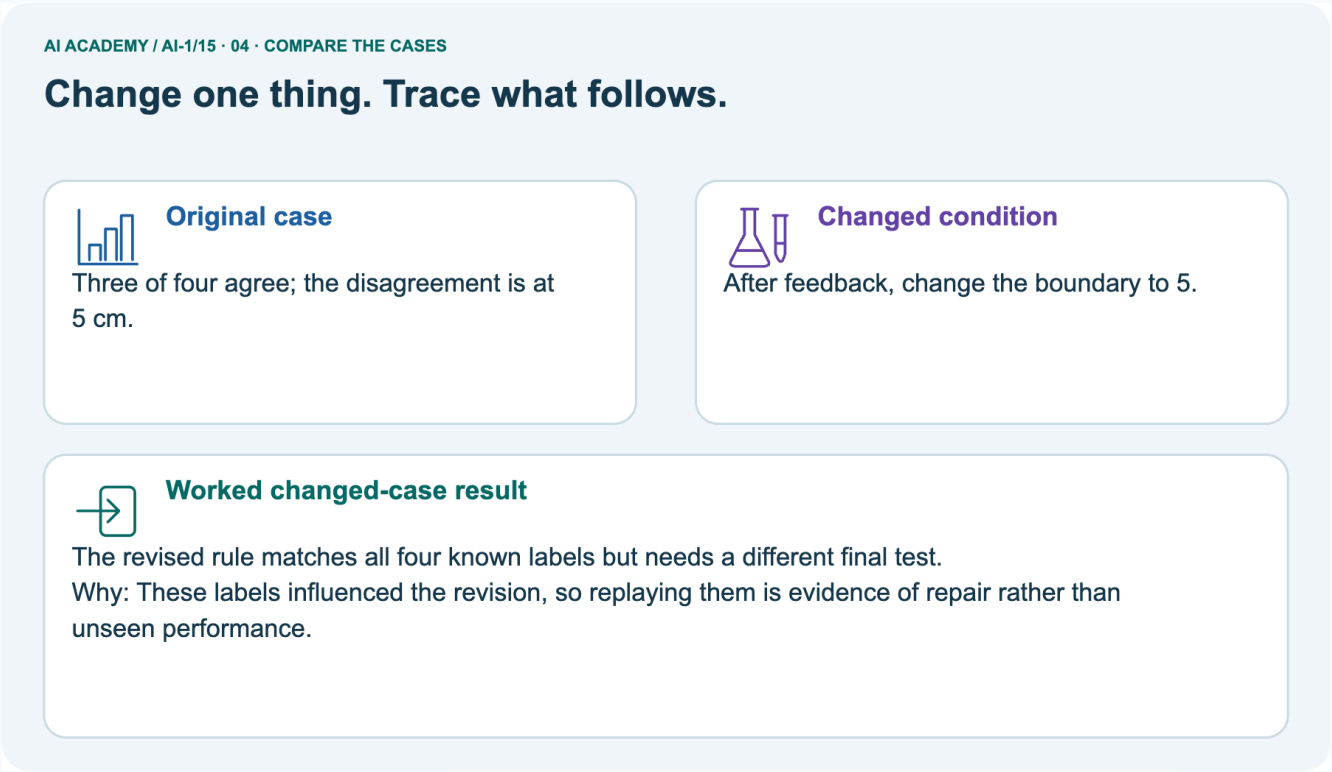
Frozen **prediction**s versus reserved labels. Test 1: Length cm = 3, PencilP = short, Reference = short, Result = agree. Test 2: Length cm = 5, PencilP = short, Reference = long, Result = disagree.

Test 3: Length cm = 6, PencilP = long, Reference = long, Result = agree. Test 4: Length cm = 9, PencilP = long, Reference = long, Result = agree.

Read the labels and values. Explain the relationship the model shows.

My picture explanation, deciding evidence and one limit

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

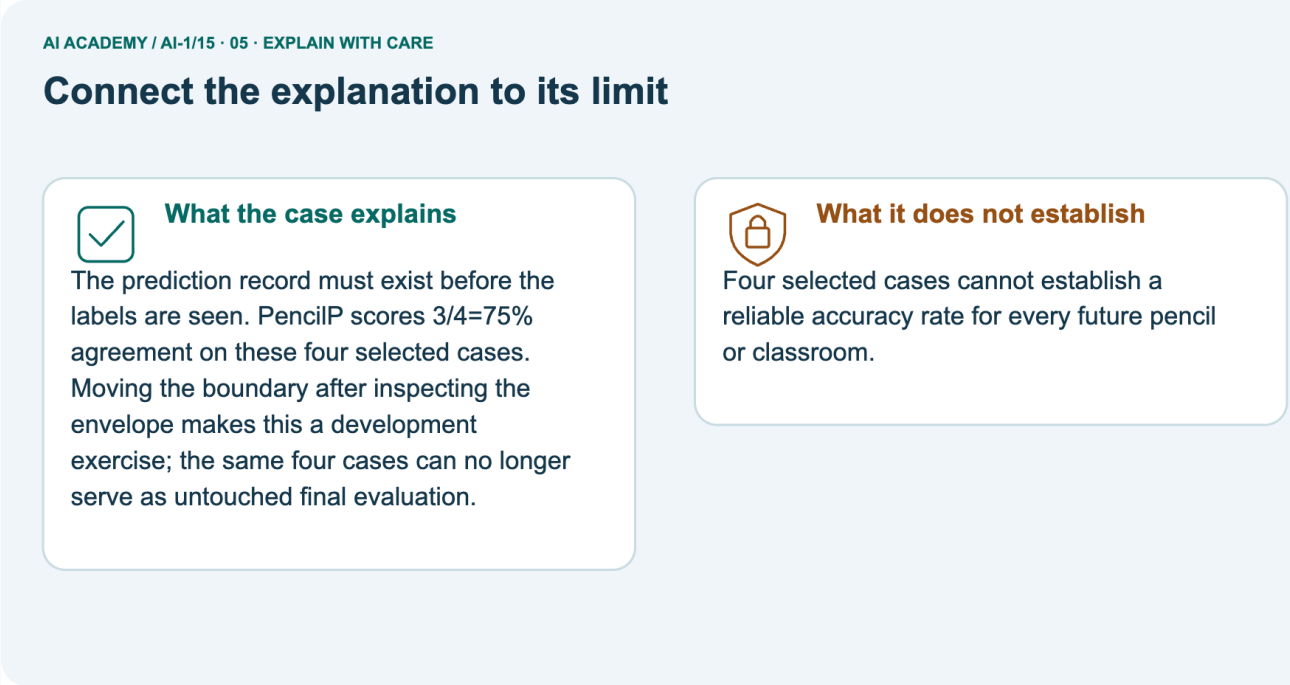
Original case: Three of four agree; the disagreement is at 5 cm. Changed condition: After feedback, change the boundary to 5. Expected result: The revised rule matches all four known labels but needs a different final test.

Reason: These labels influenced the revision, so replaying them is **evidence** of repair rather than unseen performance.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

My picture explanation, deciding evidence and one limit

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

The **prediction** record must exist before the labels are seen. PencilP scores 3/4=75% agreement on these four selected cases.

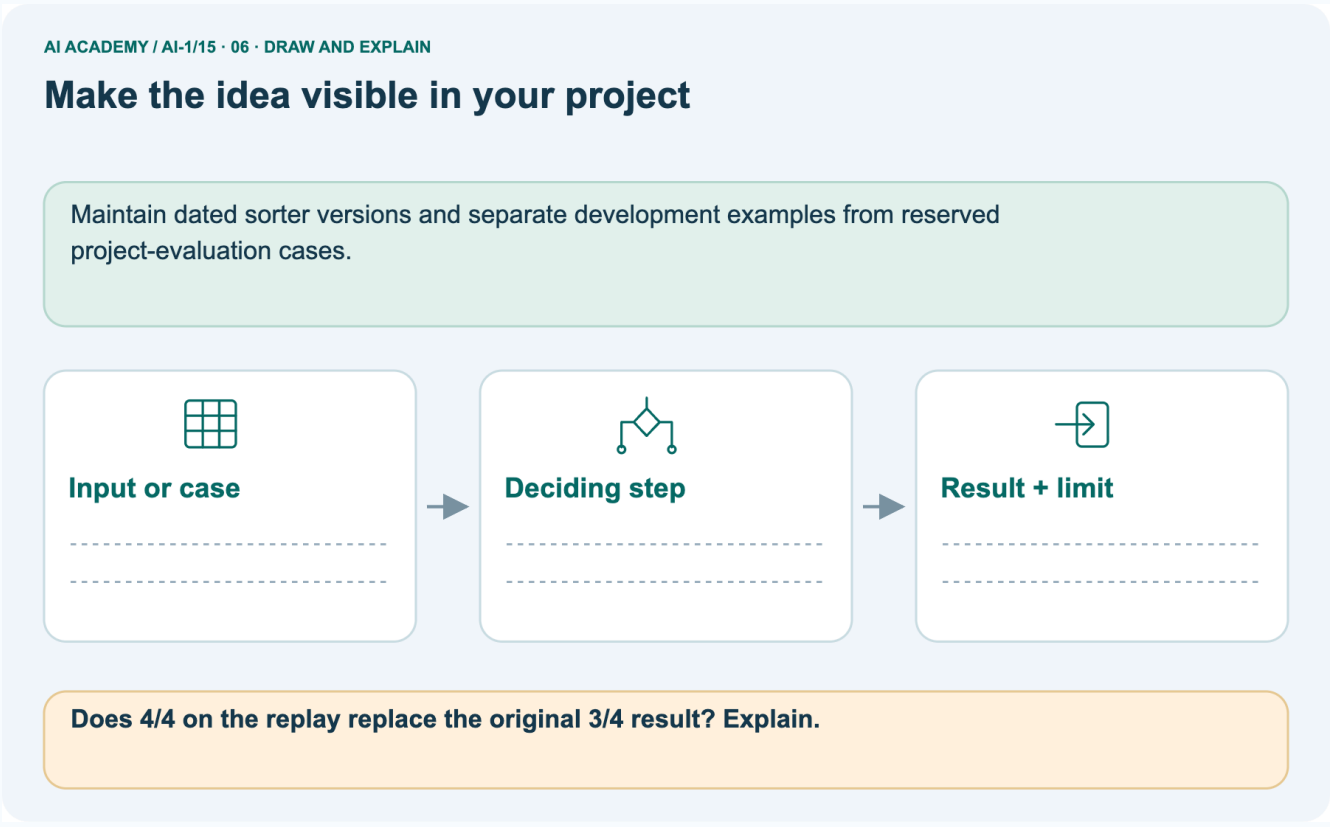
Moving the boundary after inspecting the envelope makes this a development exercise; the same four cases can no longer serve as untouched final evaluation.

Limit: Four selected cases cannot establish a reliable accuracy rate for every future pencil or classroom.

> Explain the case, then point to a limit on the claim.

My picture explanation, deciding evidence and one limit

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

Maintain dated sorter versions and separate **development examples** from reserved project-evaluation cases. Draw your **input**, deciding step, result and limit.

Supported question: Does 4/4 on the replay replace the original 3/4 result? Explain.

Draw your project trace and justify the deciding step using **evidence**.

My picture explanation, deciding evidence and one limit

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.